In [29]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Load Dataset

In [30]:
```
df = pd.read_csv("GlobalWeatherRepository.csv")
df.head()
```

Out[30]:

	country	location_name	latitude	longitude	timezone	last_updated_epoch	las
0	Afghanistan	Kabul	34.52	69.18	Asia/Kabul	1715849100	2
1	Albania	Tirana	41.33	19.82	Europe/Tirane	1715849100	2
2	Algeria	Algiers	36.76	3.05	Africa/Algiers	1715849100	2
3	Andorra	Andorra La Vella	42.50	1.52	Europe/Andorra	1715849100	2
4	Angola	Luanda	-8.84	13.23	Africa/Luanda	1715849100	2

5 rows × 41 columns

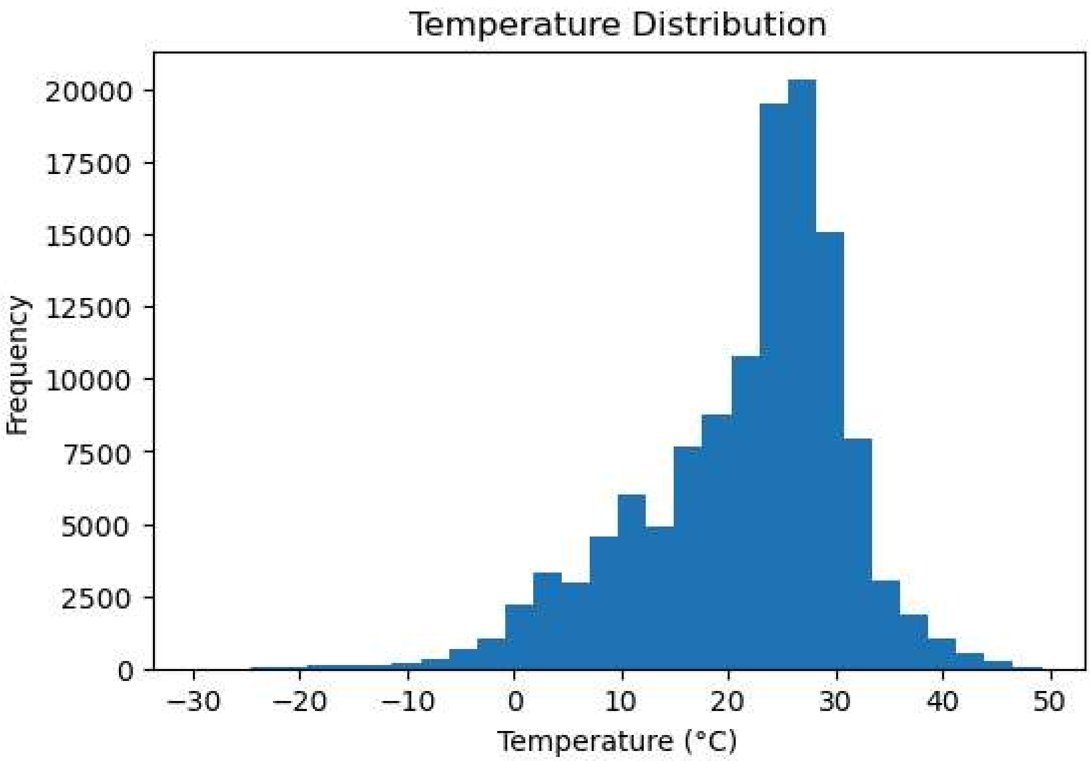
In [31]:
```
df['last_updated'] = pd.to_datetime(df['last_updated'], errors='coerce')

# Extract time features
df['year'] = df['last_updated'].dt.year
df['month'] = df['last_updated'].dt.month
```

In [32]:
```
print("\nStatistical Summary:")
print(df[['temperature_celsius','humidity','precip_mm','wind_kph']].describe())
```

```
Statistical Summary:
       temperature_celsius      humidity     precip_mm      wind_kph
count        123356.000000  123356.000000  123356.000000  123356.000000
mean             21.665486      66.061383      0.137063      12.995777
std               9.547590      24.028847      0.572792      11.891706
min             -29.800000       2.000000      0.000000       3.600000
25%              16.300000      50.000000      0.000000       6.500000
50%              24.100000      71.000000      0.000000      11.200000
75%              28.100000      85.000000      0.030000      17.600000
max              49.200000     100.000000     42.240000    2963.200000
```

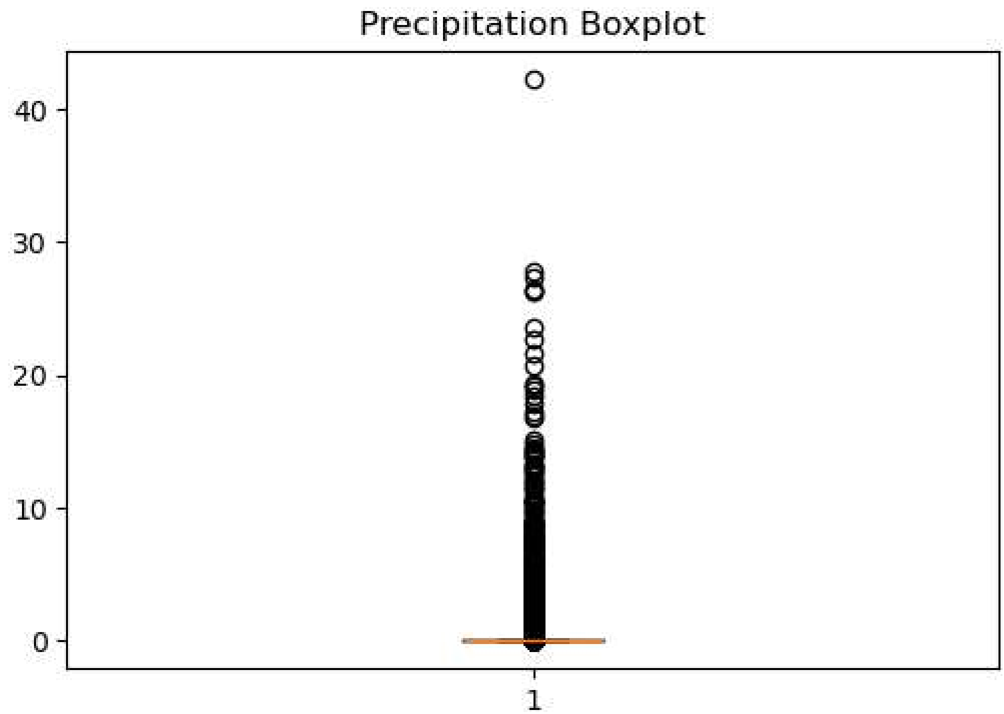
Distribution Analysis

In [33]:
```
plt.figure(figsize=(6,4))
plt.hist(df['temperature_celsius'], bins=30)
plt.title("Temperature Distribution")
plt.xlabel("Temperature (°C)")
plt.ylabel("Frequency")
plt.show()
```



Boxplot

In [34]:
```
plt.figure(figsize=(6,4))
plt.boxplot(df['precip_mm'])
plt.title("Precipitation Boxplot")
plt.show()
```



Correlation Matrix

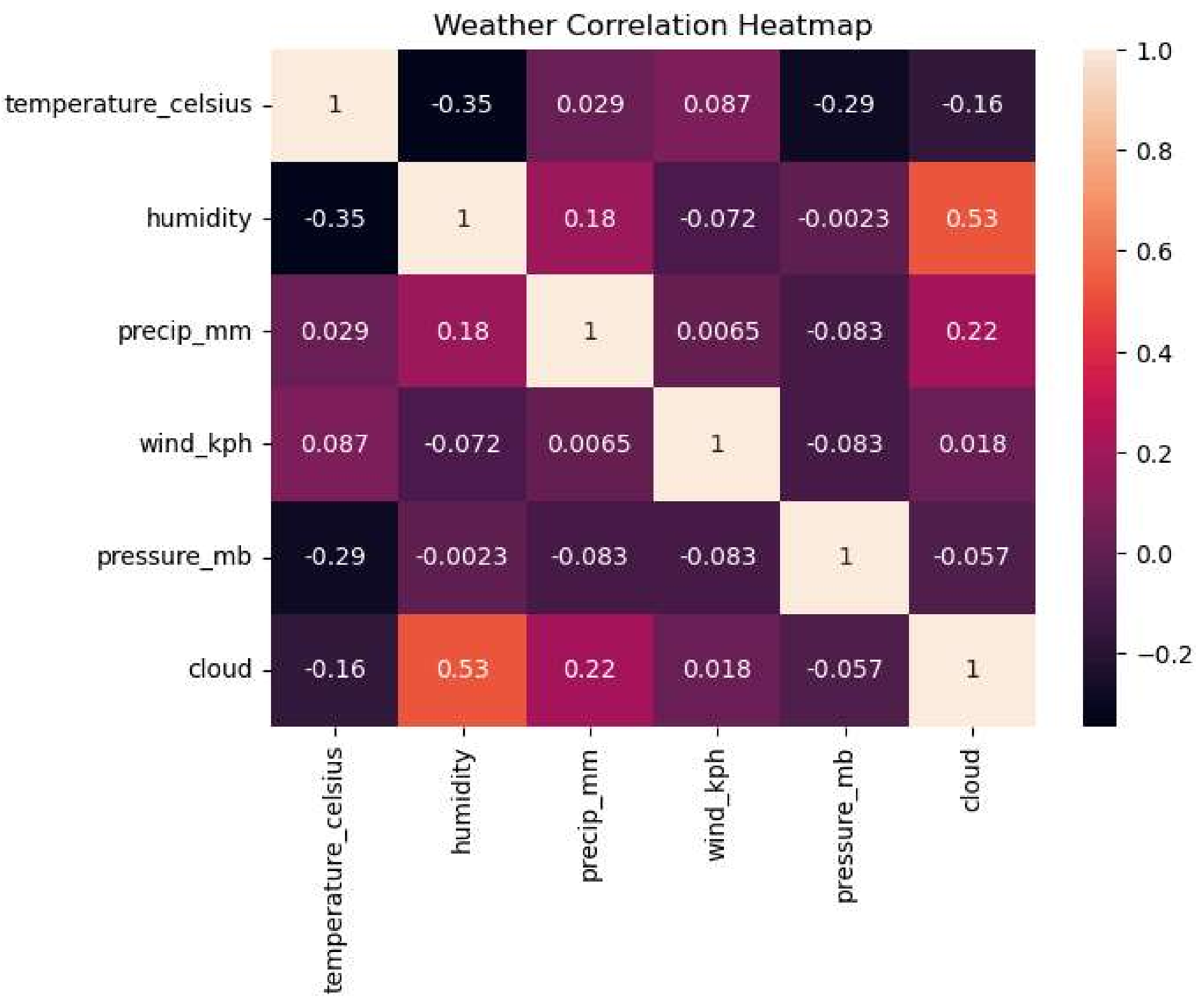
In [38]:
```
corr = df[corr_cols].corr()
print(corr)
```

```
                    temperature_celsius  humidity  precip_mm  wind_kph  \
temperature_celsius            1.000000 -0.345047   0.029176  0.086903
humidity                      -0.345047  1.000000   0.175261 -0.072008
precip_mm                      0.029176  0.175261   1.000000  0.006481
wind_kph                       0.086903 -0.072008   0.006481  1.000000
pressure_mb                   -0.287561 -0.002291  -0.082704 -0.082765
cloud                         -0.157217  0.525465   0.215279  0.018473

                    pressure_mb     cloud
temperature_celsius   -0.287561 -0.157217
humidity              -0.002291  0.525465
precip_mm             -0.082704  0.215279
wind_kph              -0.082765  0.018473
pressure_mb            1.000000 -0.056716
cloud                 -0.056716  1.000000
```

Correlation Heatmap

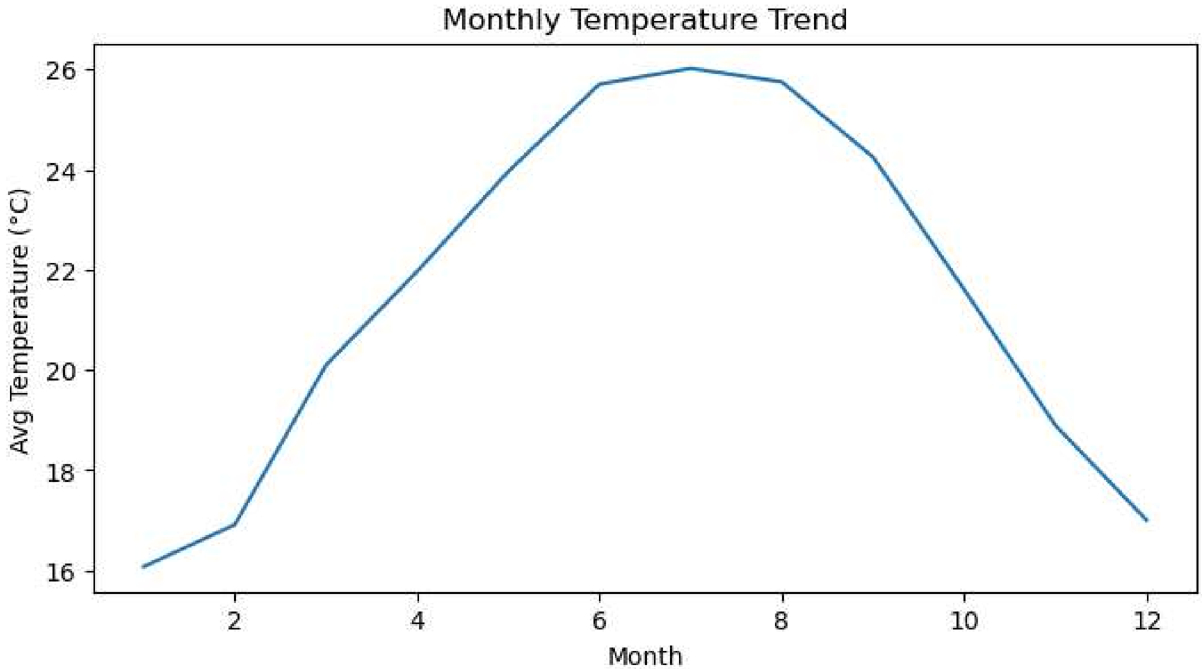
In [40]:
```
corr_cols = [
    'temperature_celsius',
    'humidity',
    'precip_mm',
    'wind_kph',
    'pressure_mb',
    'cloud'
 ]
```

```
corr = df[corr_cols].corr()

plt.figure(figsize=(7,5))
sns.heatmap(corr, annot=True)
plt.title("Weather Correlation Heatmap")
plt.show()
```



Seasonal Trend

In [41]:
```
monthly_temp = df.groupby('month')['temperature_celsius'].mean()

plt.figure(figsize=(8,4))
plt.plot(monthly_temp.index, monthly_temp.values)
plt.title("Monthly Temperature Trend")
plt.xlabel("Month")
plt.ylabel("Avg Temperature (°C)")
plt.show()
```



Extreme Weather Detection

In [43]:
```
mean_temp = df['temperature_celsius' ].mean ()
std_temp = df ['temperature_celsius' ] . std ()

high_threshold = mean_temp + 2 * std_temp
low_threshold = mean_temp - 2 * std_temp

extreme_temp = df[
(df['temperature_celsius' ] > high_threshold) |
(df['temperature_celsius' ] < low_threshold)

]

extreme_temp[ ['country', 'temperature_celsius' ]]
```

Out[43]:

	country	temperature_celsius
35	Chile	1.0
77	India	42.0
229	Chad	42.0
230	Chile	2.0
346	Saudi Arabia	41.0
...	...	...
123314	Serbia	1.1
123329	Sweden	-7.9
123330	Switzerland	0.4
123336	Bulgaria	-3.8
123344	Ukraine	-15.7

6811 rows × 2 columns

Extreme Rainfall

In [44]:

```
mean_rain = df['precip_mm' ].mean ()
std_rain = df['precip_mm' ]. std()

rain_threshold = mean_rain + 2 * std_rain

extreme_rain = df[df['precip_mm' ] > rain_threshold]

extreme_rain[['country', 'precip_mm' ]]
```

Out[44]:

	country	precip_mm
103	Malaysia	1.86
135	Papua New Guinea	1.79
156	Singapore	1.75
176	Tonga	2.09
181	Tuvalu	2.00
...	...	...
123140	Timor-Leste	2.22
123164	Andorra	2.39
123296	Papua New Guinea	1.69
123301	Portugal	1.41
123320	Solomon Islands	1.60

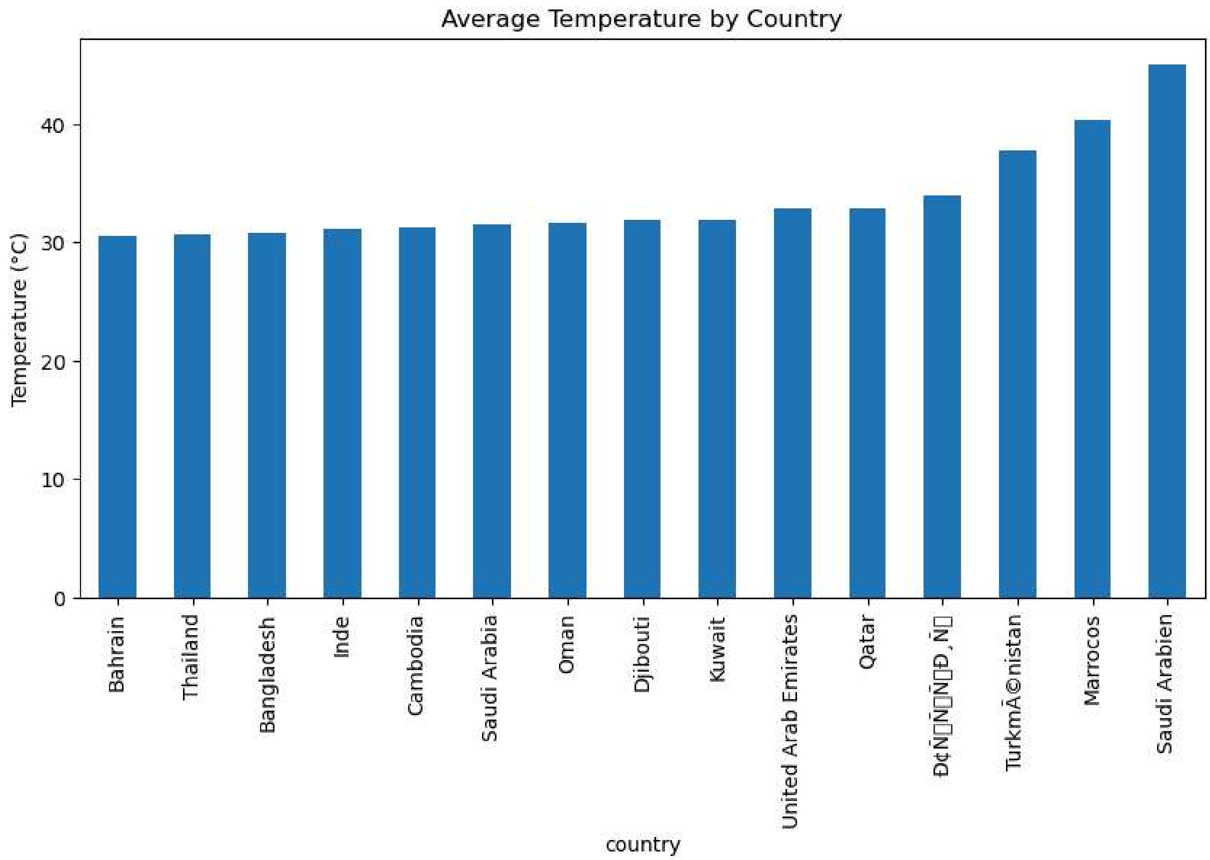
3308 rows × 2 columns

Regional Comparison

In [45]:

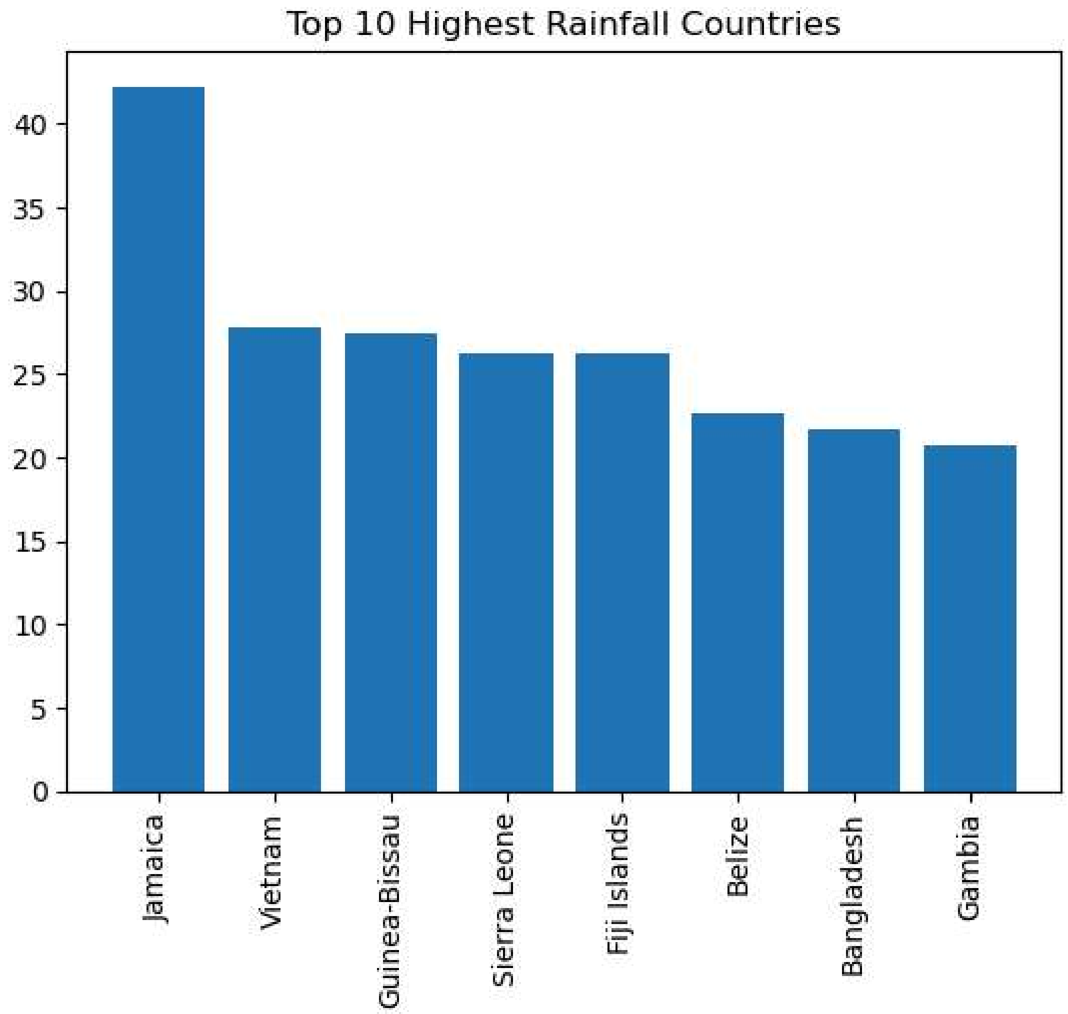
```
regional_temp = df.groupby('country')['temperature_celsius'].mean().sort_values()

plt.figure(figsize=(10,5))
regional_temp.tail(15).plot(kind='bar')
plt.title("Average Temperature by Country")
plt.ylabel("Temperature (°C)")
plt.show()
```

```
C:\Users\harik\anaconda3\Lib\site-packages\IPython\core\pylabtools.py:170: UserWarni
ng: Glyph 131 (\x83) missing from font(s) DejaVu Sans.
  fig.canvas.print_figure(bytes_io, **kw)
C:\Users\harik\anaconda3\Lib\site-packages\IPython\core\pylabtools.py:170: UserWarni
ng: Glyph 128 (\x80) missing from font(s) DejaVu Sans.
  fig.canvas.print_figure(bytes_io, **kw)
C:\Users\harik\anaconda3\Lib\site-packages\IPython\core\pylabtools.py:170: UserWarni
ng: Glyph 134 (\x86) missing from font(s) DejaVu Sans.
  fig.canvas.print_figure(bytes_io, **kw)
C:\Users\harik\anaconda3\Lib\site-packages\IPython\core\pylabtools.py:170: UserWarni
ng: Glyph 143 (\x8f) missing from font(s) DejaVu Sans.
  fig.canvas.print_figure(bytes_io, **kw)
```



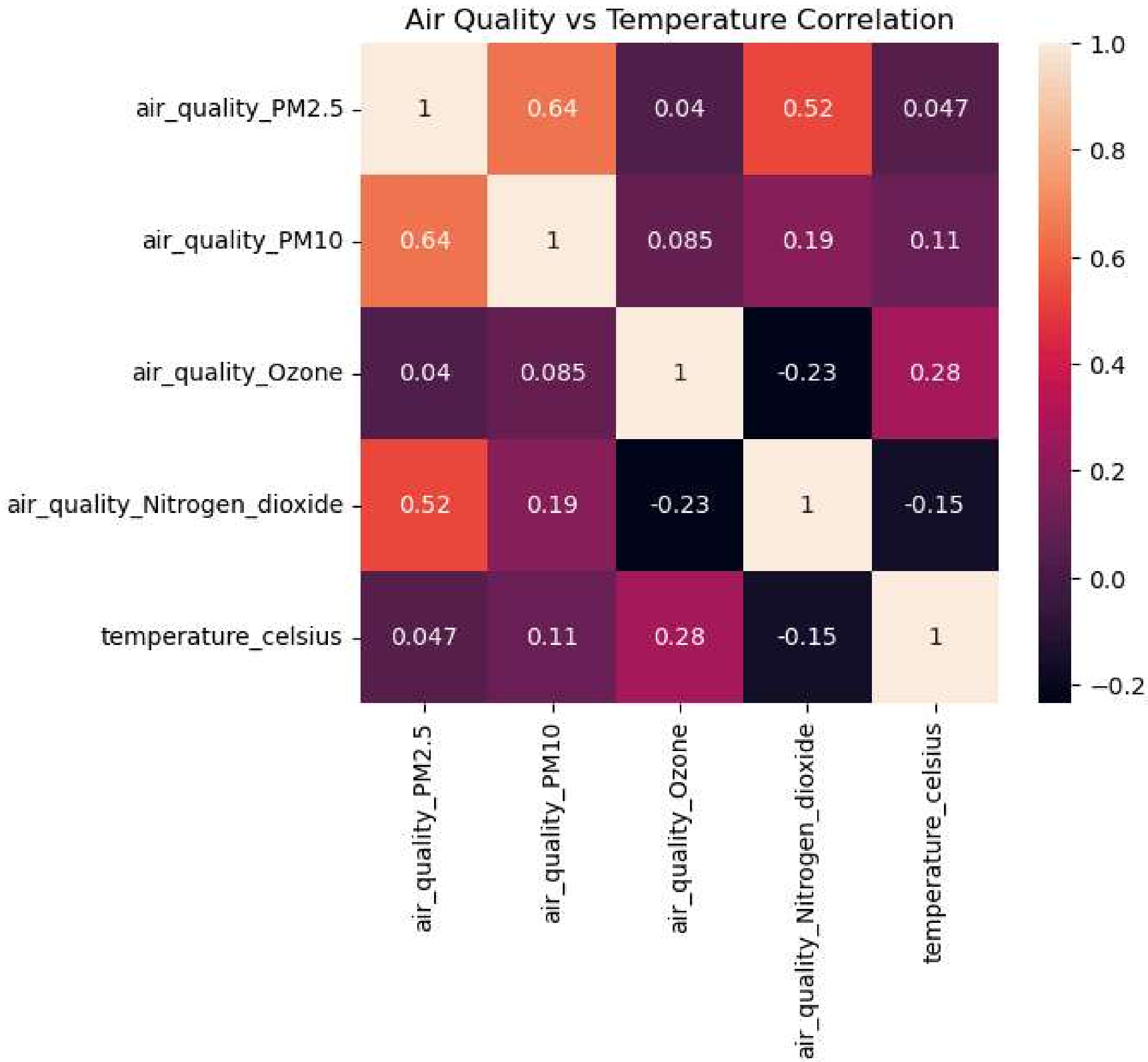
Top 10 Rainfall Countries

In [50]:
```
top_rain = df.sort_values ('precip_mm', ascending=False) .head(10)

plt.figure()
plt.bar(top_rain['country' ], top_rain['precip_mm' ])
plt.xticks(rotation=90)
plt.title("Top 10 Highest Rainfall Countries")
plt.show()
```



Air Quality

In [47]:
```
pollution_cols = [
    'air_quality_PM2.5',
    'air_quality_PM10',
    'air_quality_Ozone',
    'air_quality_Nitrogen_dioxide'
]
corr_pollution = df[pollution_cols + ['temperature_celsius' ]]. corr()

plt.figure(figsize=(6,5))
sns.heatmap(corr_pollution, annot=True)
plt.title("Air Quality vs Temperature Correlation")
plt.show()
```


Air Quality vs Temperature Correlation
air_quality_PM2.5
air_quality_PM10
air_quality_Ozone
air_quality_Nitrogen_dioxide
temperature_celsius
1 0.64 0.04 0.52 0.047
0.64 1 0.085 0.19 0.11
0.04 0.085 1 -0.23 0.28
0.52 0.19 -0.23 1 -0.15
0.047 0.11 0.28 -0.15 1
1.0
0.8
0.6
0.4
0.2
0.0
−0.2

```
In [ ]:
```